ModelIt!

Teacher's Guide

Is Our Freshwater Running Out?

Modeling the Water Cycle, Aquifer Depletion, and Global Water Security

Quick Reference

Grade Level	Grade 12L2
NGSS Standard	HS-ESS2-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS2-5

Plan and conduct an investigation of the properties of water and its effects on Earth materials and surface processes.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students investigate the properties of water systems by modeling how precipitation, extraction, and recharge interact to determine groundwater availability, and how changing one variable propagates through the interconnected system.
Disciplinary Core Idea	**ESS2.C The Roles of Water in Earth\'s Surface Processes** — Water participates in both the chemical and physical processes that shape Earth\'s surface, including the formation and depletion of groundwater systems that are critical resources for human civilization.
Crosscutting Concept	**Cause and Effect** — Students trace the causal chains from human water extraction and climate change through groundwater depletion, ecosystem degradation, and water quality decline, identifying how short-term water use decisions produce long-term irreversible consequences.

CAST Testing Alignment

MULTIPLE CHOICE:

An aquifer has been declining 1.5 meters per year for 20 years. Climate models project 15% less precipitation in the region over the next 30 years. Which best describes the expected change to the depletion timeline? A) The aquifer will deplete at the same rate because pumping is constant B) Depletion will accelerate because reduced precipitation decreases recharge while demand stays the same C) Depletion will slow because drought reduces agricultural demand D) The aquifer will stabilize as the water table drops below economical pumping depth

CONSTRUCTED RESPONSE:

Using your water system model, explain why a region that improves irrigation efficiency by 40% may still face groundwater depletion if population grows by 30%. Reference the concepts of recharge rate, extraction rate, and water balance in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Water is the most essential resource for human civilization, yet it is among the most poorly managed. While 71% of Earth\'s surface is covered by water, 97.5% is saltwater. Of the remaining 2.5% freshwater, most is locked in glaciers and ice caps. Only about 0.5% of Earth\'s total water is accessible freshwater in rivers, lakes, and usable aquifers. This finite supply is under growing pressure from population growth, agricultural expansion, industrial demand, and climate change — a combination that is depleting aquifers faster than they can recharge and threatening the water security of billions.

The Groundwater Crisis

Groundwater stored in aquifers provides drinking water for over 2 billion people and irrigates 40% of the world\'s food crops. Many of the world\'s most important aquifers are being depleted at alarming rates. The Ogallala Aquifer under the US Great Plains — which irrigates one-third of America\'s cropland — has lost 30% of its volume since 1960, with some areas losing over 3 meters of water level per year. The Indo-Gangetic aquifer system that supports 600 million people in India, Pakistan, and Bangladesh is declining even faster. NASA\'s GRACE satellites have documented significant groundwater depletion in 21 of the world\'s 37 largest aquifer systems. The fundamental problem is a timescale mismatch: these aquifers were filled over thousands to millions of years through slow geological recharge, but modern pumping technology can drain them in decades.

Agriculture: The Thirsty Giant

Agriculture accounts for approximately 70% of global freshwater withdrawals, making it by far the largest consumer. Much of this water is used inefficiently: flood irrigation, which simply inundates fields, loses 40-60% of water to evaporation and runoff. Switching to drip irrigation can reduce water use by 50-70% while actually increasing crop yields. However, most of the world\'s irrigated farmland still uses flood methods because drip systems require capital investment that small-scale farmers cannot afford. The types of crops matter enormously: producing 1 kilogram of wheat requires about 1,500 liters of water, while 1 kilogram of beef requires about 15,000 liters (because cattle consume water-intensive grain over their lifetime). Dietary choices and agricultural efficiency are therefore the largest levers for reducing water demand.

Climate Change and Water: The Double Squeeze

Climate change affects water availability through two simultaneous mechanisms. First, precipitation patterns are shifting: wet regions are generally getting wetter while dry regions are getting drier, intensifying existing water stress. Second, higher temperatures increase evapotranspiration — the rate at which water evaporates from soil and transpires from plants — meaning that even with the same precipitation, less water is available because more is lost to the atmosphere. Climate models project that regions already experiencing water stress — the Middle East, North Africa, Central Asia, southwestern US, and southern Australia — will face 10-30% less water availability by 2050. Simultaneously, melting glaciers that provide seasonal meltwater for billions of people in South America and South Asia will lose 30-60% of their volume, disrupting river flows that are critical for drinking water and irrigation.

Solutions: From Efficiency to Desalination

Water management solutions span a spectrum of cost and complexity. The cheapest interventions are efficiency improvements: drip irrigation, leak repair in distribution systems (many cities lose 30-50% of treated water to leaks), and water-efficient appliances. Water recycling for industrial and agricultural use can return 60-90% of wastewater to productive use. Managed aquifer recharge — intentionally directing excess surface water underground during wet periods — can partially offset depletion. Desalination can convert seawater to freshwater but requires significant energy (3-10 kWh per cubic meter), produces concentrated brine that harms marine ecosystems when discharged, and costs 3-5 times more than conventional freshwater in most locations. Israel has demonstrated that combining aggressive efficiency, recycling (87% of wastewater is

recycled for agriculture), and desalination can transform a desert nation into a water-secure one, but this model requires sustained investment and political commitment.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Precipitation Rate, Population and Agricultural Demand, Surface Water Availability, Groundwater Level, Water Recycling and Efficiency, Ecosystem Water Requirements, and Water Quality Index as key components of the water system.

E **Establish Relationships**

Students determine that extraction exceeds recharge in most agricultural aquifers, that climate change reduces both precipitation and effective water availability, that agriculture dominates demand, and that efficiency and recycling can partially but not fully offset growing demand.

V **Visualize & Model**

Students build a computational model of a regional water system showing how precipitation, extraction, and conservation measures determine the long-term trajectory of groundwater levels and water availability.

E **Evaluate Outcomes**

Students run increasing extraction, climate-driven drought, and efficiency-and-reuse scenarios to test predictions about water system sustainability and the effectiveness of different management strategies.

R **Revise & Extend**

Students add virtual water trade, desalination infrastructure, or managed aquifer recharge to explore how additional factors affect regional water security and the options for addressing water stress.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Water covers most of our planet, but only 0.5% is freshwater we can actually use. Today we\'re modeling why that tiny fraction is running out in the places that need it most."*

Watch For

Show cover image. Quick stat: 2 billion people already live in water-stressed regions. Ask: What do you think would happen if the aquifer under America\'s breadbasket ran dry?

Slide 2: Learning Objectives

3 min

Say: *"We\'re modeling the complete water system — from rain falling to aquifer depletion. By the end, you\'ll understand why water crises are coming and what can be done to prevent them."*

Watch For

Have students read objectives. Pre-teach vocabulary. Key concept: the timescale mismatch — aquifers fill over thousands of years but can be drained in decades. This is the core problem.

Slide 3: Big Question

3 min

Say: *"We\'re draining aquifers that took thousands of years to fill. What happens to the 2 billion people and the agriculture that depends on them when the water is gone?"*

Watch For

Quick-write: Students predict what happens to a farming region when its aquifer is depleted. What are the consequences for food production, communities, and ecosystems?

Slide 4: LEVER Framework

2 min

Say: *"We\'ll model a regional water system to understand the balance between recharge and extraction. This model will reveal timelines that most people don\'t realize."*

Watch For

Preview LEVER steps. Emphasize that modeling reveals hidden timelines — the water crisis isn\'t coming someday, it\'s arriving on a predictable schedule that the model can calculate.

Slide 5: Activity 1: Components

8 min

Say: *"Which components are external inputs to the water system, and which are internal responses? Think about what drives water availability versus what responds to those drivers."*

Watch For

Guide sorting: External = Precipitation Rate (climate), Population and Agricultural Demand (human choices). Internal = the five water system responses. Discuss: Why is Groundwater Level internal when it seems like a fixed resource?

Slide 6: Activity 2: Connections

8 min

Say: *"When demand exceeds what precipitation and surface water can provide, where does the extra water come from? What happens to that source over time? Trace the pathway to crisis."*

Watch For

Students draw arrows. Key insight: groundwater acts as a buffer — it fills the gap between demand and renewable supply. But unlike a bank account with deposits, the recharge deposits are tiny compared to the withdrawals. Ask: What happens when the buffer runs out?

Slide 7: Activity 3: Simulation

10 min

Say: *"Run all three scenarios and focus on the groundwater level graph. In which scenario does the aquifer stabilize? In which does it hit zero? What makes the difference?"*

Watch For

Students run scenarios. Key discussion: Compare the depletion timelines across scenarios. Even the efficiency scenario only delays depletion — it doesn\'t prevent it unless demand growth is also addressed. Ask: What would it take to actually stabilize the aquifer?

Slide 8: Discoveries

5 min

Say: *"The most important number: aquifers took thousands of years to fill and are being drained in decades. This isn\'t a water cycle problem — it\'s a timescale mismatch between geology and technology."*

Watch For

Class discussion. Compare predictions to model evidence. Challenge question: If we know the aquifer will run out in 25 years, why isn\'t that causing an emergency response right now? What explains the gap between knowledge and action?

Slide 9: STEM Challenge

12 min

Say: *"Your aquifer is declining 1.5 meters per year and will be too depleted to pump within 25 years. Design a water system that stabilizes the aquifer while keeping agriculture productive and the population growing. Every decision must be justified by your model data."*

Watch For

Groups design comprehensive water management plans addressing efficiency, recycling, alternative sources, and allocation governance. Present with specific numbers from model simulations.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Precipitation Rate, Population and Agricultural Demand
- **INTERNAL:** Surface Water Availability, Groundwater Level, Water Recycling and Efficiency, Ecosystem Water Requirements, Water Quality Index

Relationships

- **Precipitation Rate to Surface Water Availability:** POSITIVE (+) — Higher precipitation increases river flows, lake levels, and reservoir storage, making more surface water available for human use and ecosystem needs.
- **Precipitation Rate to Groundwater Level:** POSITIVE (+) — Precipitation contributes to aquifer recharge through percolation, though the recharge rate is very slow compared to extraction — only a small fraction of precipitation actually reaches the aquifer.
- **Population and Agricultural Demand to Groundwater Level:** NEGATIVE (-) — Higher demand increases groundwater pumping. When extraction exceeds natural recharge, the aquifer depletes progressively, with the rate of decline proportional to the gap between extraction and recharge.
- **Groundwater Level to Surface Water Availability:** POSITIVE (+) — Groundwater and surface water are connected — healthy aquifers feed springs and baseflow to rivers during dry periods. As aquifer levels drop, springs dry up and river baseflows decline, reducing surface water availability.
- **Water Recycling and Efficiency to Groundwater Level:** POSITIVE (+) — Better efficiency and recycling reduce the total volume of water that must be extracted from the aquifer, slowing depletion. Drip irrigation is the single largest efficiency opportunity because agriculture dominates demand.
- **Ecosystem Water Requirements to Water Quality Index:** POSITIVE (+) — When ecosystems receive sufficient water to maintain healthy rivers and wetlands, these ecosystems provide natural water purification services. When environmental flows are sacrificed to meet human demand, water quality declines as these natural treatment systems degrade.

Key Discoveries

- Groundwater depletion is effectively irreversible on human timescales — the model shows that aquifers being drained in decades took thousands of years to fill through natural recharge, meaning once depleted, they cannot be replenished within any relevant planning horizon
- Agriculture is the dominant driver of water stress, consuming 70% of freshwater — the model shows that even modest improvements in irrigation efficiency (switching from flood to drip irrigation) have larger impacts on the water balance than all other demand reduction measures combined
- Climate change acts as a threat multiplier by simultaneously reducing precipitation in already dry regions and increasing evapotranspiration through higher temperatures — the combination compresses the timeline for water crisis compared to either factor alone
- Water recycling and efficiency improvements can offset 30-40% of growing demand, but in regions where demand already exceeds sustainable supply, efficiency alone is insufficient — structural changes to water allocation and agricultural practices are required

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The water cycle means water is always renewable"	The water cycle does continuously recycle water through evaporation, precipitation, and runoff. However, groundwater operates on a very different timescale from surface water. While rain and river water cycle in days to months, groundwater in deep aquifers can be thousands to millions of years old and recharges at millimeters per year. Extracting this ancient groundwater faster than it recharges is functionally equivalent to mining a non-renewable resource. The water cycle does not replenish aquifers on any timescale relevant to human civilization once they are depleted.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Desalination can solve all water scarcity problems"	While desalination technology works and is improving, it faces significant limitations. It requires 3-10 kWh per cubic meter of freshwater produced — enormous energy demand at scale. It produces concentrated brine waste that harms marine ecosystems when discharged. It costs 3-5 times more than conventional freshwater in most locations. And it is only practical for coastal areas — landlocked regions and their agriculture cannot benefit. Desalination is one tool in the water management toolkit, but it is not a universal solution and cannot replace sustainable management of existing freshwater resources.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Water scarcity is a developing-world problem that doesn\'t affect wealthy nations"	The western United States, Australia, and southern Europe all face severe water stress. The Colorado River — which supplies water to 40 million Americans — no longer reaches the sea because every drop is extracted upstream. California\'s Central Valley has sunk by up to 9 meters due to groundwater depletion. Cape Town, South Africa nearly ran out of water in 2018. Wealthy nations can afford expensive infrastructure, but they cannot create water that doesn\'t exist. As climate change intensifies drought and aquifers continue declining, water scarcity will increasingly affect rich and poor nations alike.